1 STRATA Benchmark Construction

1.1 Annotated timelines
(MedTempo)

3,166 VAERS reports, three vaccines, expert-annotated as a leveled order of events

Example timeline
report MT-0482913



stage → events; the first mention fixes the stage; events at one stage co-exist

1.2 Hypothesis templates

Precedes	A tends to precede B
Co-occurs	A and B tend to co-occur
Sequence	A, then B, then C
First / Last	A is most often first (last)
Position	A falls in the earlier half
Proximity	A occurs right before B
Threshold	A precedes B in ≥ 2/3

1.3 Groups, families, levels

26 groups
9 report, 16 cohort, 1 two-cohort

7 claim families
precedes, co-occurs, sequence, first/last, position, proximity, threshold

3 difficulty levels
Easy one report; Medium one cohort; Hard two cohorts

1.4 Gold annotations

Plan
events, quantity, denominator primitive, arms, null, p0, tail, test, floor

Evidence sets
denominator, numerator and excluded ids; n, k per arm

Statistics, verdict
p on both tails, Wilson CI, Cohen's h; supported / refuted / insufficient

validated by invariants and an oracle; permutation-screened

2 ART Verification Pipeline

Inputs

a claim in natural language, with its cohort and level

Resources (to S1)

codebook: the corpus stipulations as text; null sources; test definitions

annotation DB: report, vaccine, term, first-mention stage

S1 LLM — Articulate
form a structured plan; the claim-type mapping is withheld

1. **Parse the claim** — events, cohort, quantity, direction
2. **Choose the primitive** — denominator or rule; arms
3. **Compose the plan** — null, p0, tail, test family, floor
4. **Verify, 1 round** — plan vs the codebook; re-ask once

Contract: the harness resolves the plan's primitive into its exact rule and states it to S2

S2 LLM — Retrieve
plan to evidence sets

told the schema and the contract; one SQL per arm, one row per report

sandbox: read-only execution → n, k and report ids per arm

verify, up to 2 rounds: rejected SQL, n = k, empty arm, rule leaks

S3 function — Test
exact, deterministic

exact binomial vs p0; Fisher's exact test for two arms; modal-vs-runner-up for argmax; floor n ≥ 10

- **Supported**
- **Refuted**
- **Insufficient**

3 Stage-level Evaluation

any stage can be set to gold

Plan correctness
plan EM, claim-type EM

Evidence correctness
evidence F1, co-exist / tie leak, faithfulness

Verdict and abstention
accuracy, macro P/R/F1, overclaim, underclaim